

MATH 6388: Statistical and Machine Learning

Course Syllabus

Course Information

- **Instructor:** Farhad Pourkamali, Ph.D., Associate Professor
- **Email:** farhad.pourkamali@ucdenver.edu
- **Office Hours:** Mo 12:00PM-1:15PM and We 3:15PM-4:00PM
 - Office: Student Commons Building (SCB) 4215
 - If you cannot make the regularly scheduled office hours, email me to arrange a different time that works for both of us. We can also meet by Zoom.

Course Description

Statistical and machine learning focuses on developing mathematical models that learn patterns from data to make predictions, quantify uncertainty, and support data-driven decision-making. This course provides a mathematically rigorous introduction to modern statistical and machine learning, emphasizing the fundamental principles that unify a broad range of learning algorithms. Throughout the semester, students develop the mathematical foundations of the field, including multivariable calculus, linear algebra, probability, statistics, and optimization. Particular emphasis is placed on the statistical aspects of learning, including probabilistic modeling, the relationship between statistical assumptions and loss function selection, parameter estimation, predictive uncertainty, calibration, and reliability. Students also gain practical experience implementing these concepts using Python and PyTorch.

Building upon these foundations, the course progresses from computational aspects of machine learning, including tensors, automatic differentiation, computational graphs, and gradient-based optimization, to regression and classification models. The course then introduces modern deep learning architectures for image and sequential data, including convolutional and recurrent neural networks, before concluding with self-attention, Transformer architectures, Vision Transformers, and foundation models. Therefore, students will develop both the theoretical understanding and practical skills needed to build reliable and trustworthy artificial intelligence systems.

Canvas and Course Announcements

The course material (e.g., syllabus, lecture notes, assignments, grades, web resources, etc.) will be provided on the Canvas course site (<http://canvas.cuonline.edu/>). It is your responsibility to know how to access this site. Contact Canvas help or me for any questions. Course announcements will often be made via the Canvas course site, and email notifications will be sent to your university email address in accordance with university policy. I will be sending out weekly announcements to keep you in the loop. These updates will cover the week's main topics, important reminders, and any relevant course news. Be sure to check your inbox and stay engaged! You are responsible for the information contained in any email messages I send you.

Student Concerns and Grievances

Students who have concerns about the course or instructor should first meet with the instructor. If the issue is not resolved, the student should next contact the Associate Chair of the Department of Mathematical and Statistical Sciences (currently Prof. Stephen Hartke, stephen.hartke@ucdenver.edu). If the issue remains unresolved, the student should then appeal to the appropriate Associate Dean of the student's home school or college.

Disability and Access

The University of Colorado Denver is committed to ensuring the full participation of all students in its programs, including students with disabilities. If you have a disability or think you have a disability and need accommodations to succeed in this course, I encourage you to contact Disability Resources and Services (DRS) and/or speak with me as soon as you can. (DRS is located in Academic Building 1, Suite 2116, and at disabilityresources@ucdenver.edu.) I am committed to providing equal access as required by federal law, and I am interested in developing strategies for your success in this course. My goal is to create a learning experience that is as accessible and inclusive as possible. I welcome discussion about individual differences in learning, encountered barriers, and ways to maximize access. Every possible effort will be made to accommodate promptly.

Tips for Success

- **USE LECTURE NOTES:** The most important strategy is to utilize lecture notes that combine math and code, provided in the form of Jupyter Notebook files (.ipynb). Thus, the first step on the road to success is to visit <https://www.anaconda.com/download/> and download the Anaconda Distribution to your personal computer. This will allow you to manage packages and environments from a desktop application. Alternatively, you can use Google Colab, which is an online Jupyter Notebook environment. It does everything a local Notebook would do, but no software installation is necessary, and you will have access to free computing resources.
- **USE OFFICE HOURS OR EMAIL ME:** If you are struggling with lecture notes, textbook material, or assignments, please use office hours or email me with questions as soon as possible. I am always happy to help. If you are unable to attend office hours, we can find another time to meet. I will also implement an early alert system to notify students who are not making adequate progress throughout the semester (e.g., missed assignments).

Learning Objectives

By the end of this course, students should be able to identify the following concepts involved in developing and deploying machine learning models:

- The mathematical and statistical foundations of machine learning, including linear algebra, multi-variable calculus, probability, statistics, and optimization.
- The assumptions underlying regression and classification models, and how these assumptions influence model selection and predictive performance.
- The statistical interpretation of loss functions and their relationship to parameter estimation, including mean squared error, cross-entropy, and quantile (pinball) loss.
- Methods for quantifying predictive uncertainty and evaluating the reliability of model predictions through calibration and performance metrics.
- The fundamental principles of deep learning architectures, including convolutional neural networks, recurrent neural networks, and Transformer-based models.
- The implementation, training, evaluation, and interpretation of machine learning models using Python and PyTorch on tabular, image, and sequential datasets.

Learning Outcomes

This course will help build on several learning outcomes including the following:

- **Critical Thinking:** Students will enhance their critical thinking by learning when and how to apply the appropriate machine learning method given a particular study and data set. Students will further analyze the usefulness of each method using evaluation metrics, whether necessary assumptions are met, and the consequences of violating those assumptions. Students will conclude their study by finding possible ways to improve performance from statistical and computational standpoints.
- **Problem Solving:** Students will develop their problem solving skills by working through many machine learning problems. Each student’s problem solving skills (mathematical and computational) will be measured by their performance on the homework, exams, and the course project. Students will develop programming skills to utilize data science libraries to solve machine learning problems.
- **Written and Oral Communication:** An essential aspect of machine learning and data science is the ability to communicate the thought behind and results of an analysis both written and orally. Students will develop and be assessed on their written communication through their homework assignments, exams, and the course project report. They will develop their oral communication through class participation and course project presentation.

Textbook (OER)

While the lecture notes are designed to be self-sufficient for your learning, we provide a “Further Reading” section after each lecture. This section includes optional materials, such as textbook excerpts and research articles, for students interested in expanding their knowledge or exploring cutting-edge developments in the field.

- “Probabilistic Machine Learning: An Introduction,” Author: Kevin Murphy <https://probml.github.io/pml-book/book1.html>

Grading Policy

Your final grade will be determined as follows:

- Homework (6 assignments, each 8%): 48%
- Take-Home Exam: 20%
- Course Project: 32%

Final letter grades will be determined by the following scale:

Percentage	Letter Grade	Percentage	Letter Grade
92 or higher	A	72 – 78	C
90 – 92	A-	70 – 72	C-
88 – 90	B+	65 – 70	D+
82 – 88	B	60 – 65	D
80 – 82	B-	55 – 60	D-
78 – 80	C+	0 – 55	F

Homework: We will have regular homework assignments consisting of derivations, proofs, and applying machine learning methods to solve real-world problems. These assignments will allow you to solidify your mathematical understanding and gain hands-on experience using popular machine learning and data science libraries. I will provide timely feedback using the Canvas course site.

Late Homework: To support student flexibility, everyone will have five “grace days” that they may use on course activities in the “Homework” category only. Submitting an assignment anytime during the day after the submission deadline counts as using a 24-hour free extension on the homework assignment. Five days may be used one at a time or in sequence as desired. If an extension of more than five calendar days is needed due to unexpected circumstances, please contact me as soon as possible.

Take-Home Exam: Towards the end of the semester (most likely, the week of November 16th), you will be given a take-home exam designed to assess your understanding of the course material and your ability to apply key concepts.

Course Project: The course project is designed to integrate the concepts covered throughout the semester through a comprehensive statistical and machine learning study. The instructor will provide a real-world prediction problem and corresponding dataset, and students will develop a complete end-to-end machine learning solution. This includes data preprocessing, implementing multiple machine learning models in Python using PyTorch, selecting appropriate loss functions and evaluation metrics, comparing model performance, analyzing predictive uncertainty and model reliability when appropriate, and interpreting the resulting predictions.

The project consists of several milestones that encourage steady progress throughout the semester, culminating in three final deliverables: (i) a Python/PyTorch implementation of the required methods, (ii) a professionally written technical report summarizing the methodology, experimental design, results, and conclusions, and (iii) a 10–15 minute recorded video presentation highlighting the problem, machine learning techniques, experimental findings, and key conclusions. Evaluation will emphasize the correctness of the implementation, the quality of the experimental methodology, the statistical analysis and interpretation of the results, and the clarity and professionalism of both the written report and recorded presentation.

Tentative Course Outline

Module 1. Foundations of Machine Learning (Weeks 1–3)

This module introduces the mathematical and computational foundations of modern machine learning. Students learn how models are represented, optimized, and implemented using PyTorch.

- Overview of machine learning
- Python and PyTorch review
- Tensors and tensor operations
- Automatic differentiation
- Computational graphs
- Gradient descent and optimization
- Building and training simple neural networks

Module 2. Regression and Predictive Uncertainty (Weeks 4–6)

This module focuses on supervised learning for continuous-valued prediction while introducing probabilistic models and methods for quantifying predictive uncertainty.

- Linear regression

- Neural networks for regression
- Regression loss functions (MSE and Pinball/Quantile loss)
- Probabilistic regression models
- Model evaluation
- Predictive uncertainty estimation

Module 3. Classification and Reliable Predictions (Weeks 7–9)

This module covers classification models and emphasizes techniques for evaluating and improving the reliability of probabilistic predictions.

- Logistic regression
- Softmax regression
- Cross-entropy loss
- Multilayer perceptrons
- Performance evaluation metrics
- Model calibration
- Reliability diagrams
- Confidence estimation and predictive reliability

Module 4. Deep Learning Architectures (Weeks 10–12)

This module introduces specialized neural network architectures designed for processing image and sequential data.

- Convolutional Neural Networks (CNNs)
- Image classification
- Transfer learning
- Recurrent Neural Networks (RNNs)
- Long Short-Term Memory (LSTM) networks
- Sequence modeling

Module 5. Transformers and Foundation Models (Weeks 13–15)

This module introduces the Transformer architecture and explores how attention mechanisms have enabled modern foundation models for language, vision, and other domains.

- Self-attention
- Scaled dot-product attention
- Multi-head attention
- Positional encoding
- Transformer encoder blocks
- Vision Transformers (ViTs)
- Foundation models and large-scale pretraining

University, college, and department policies

Academic Calendar

For university deadlines and procedures (such as the last day to withdraw from a course), please see the Academic Calendar.

<https://www.ucdenver.edu/student/registration-planning/academic-calendars>

Recording of Class Meetings

Class meetings held on or streamed over a video conferencing platform (such as Zoom, Microsoft Teams, etc) may be recorded and posted for all members of the class. Student participation and interaction may be included in the recording. If you have any concerns about this, please contact the instructor.

Diversity Statement

It is my intent that students from all diverse backgrounds and perspectives be well served by this course, that students' learning needs be addressed both in and out of class, and that the diversity that students bring to this class be viewed as a resource, strength and benefit. It is my intent to present materials and activities that are respectful of diversity: gender, sexuality, disability, age, socioeconomic status, ethnicity, race, and culture, etc. I would like to create a learning environment for my students that supports a diversity of thoughts, perspectives and experiences, and honors your identities (including race, gender, class, sexuality, religion, ability, etc). To help accomplish this:

- If you have a name and/or set of pronouns that differ from those that appear in your official records, please let me know!
- If you feel like your performance in the class is being impacted by your experiences outside of class, please don't hesitate to come and talk with me. I want to be a resource for you. Remember that you can also submit anonymous feedback (which will lead to me making a general announcement to the class, if necessary to address your concerns). If you prefer to speak with someone outside of the course, the Office of Diversity, Equity and Inclusion, is an excellent resource.
- I (like many people) am still in the process of learning about diverse perspectives and identities. If something was said in class (by anyone) that made you feel uncomfortable, including by me, please talk to me about it. (Again, anonymous feedback is always an option).

Your suggestions are encouraged and appreciated. Please let me know ways to improve the effectiveness of the course for you personally or for other students or student groups. In addition, if any of our class meetings conflict with your religious or other cultural events, please let me know so that we can make arrangements for you.

Health and Wellness

As a student, you may experience a range of challenges that can interfere with learning, such as strained relationships, traumas, increased anxiety, substance use, feeling down, difficulty concentrating, and/or lack of motivation. These mental health concerns or stressful events may diminish your academic performance and/or reduce your ability to participate in daily activities. If you or someone you know is struggling, you can find supportive campus and community resources at the Health Center at Auraria or the CU Denver Counseling Center. On weekends, holidays or after-hours you can contact the 24/7 Mental Health Crisis and Victim Assistance Line at 303-615-9999 or text Talk to 38255.

The University of Colorado Denver is committed the health and well-being of all students. We recognize that diminished mental health, including significant stress, mood changes, excessive worry, or problems with eating and/or sleeping can interfere with optimal academic performance. The source of such symptoms can be quite varied, and include experiences of trauma (such as sexual and relationship violence, stalking, discrimination, crimes, and accidents), responses to course work, family worries, loss, personal struggle, or crisis. If you or someone you know is struggling, you can find supportive campus and community resources at

<https://www.ucdenver.edu/counseling-center>

or by calling the CU Denver Counseling Center (303-315-7270) or the Health Center at Auraria (303-615-9999). On weekends, holidays or after-hours you can contact the 24/7 Mental Health Crisis and Victim Assistance Line at 303-615-9999 or text Talk to 38255.

Nondiscrimination and Sexual Misconduct

The University of Colorado Denver is committed to maintaining a positive learning, working and living environment. University policy and Title IX prohibit discrimination on the basis of race, color, national origin, sex, age, disability, pregnancy, creed, religion, sexual orientation, veteran status, gender identity, gender expression, political philosophy or political affiliation in admission and access to, and treatment and employment in, its educational programs and activities. University policy prohibits sexual misconduct, including harassment, domestic and dating violence, sexual assault, stalking, or related retaliation. If you have experienced any sort of sexual misconduct or discrimination, please visit the Office of Equity web site at <https://www.ucdenver.edu/offices/equity> to understand the resources available to you.

Please note that I am a Responsible Employee, which means that if I witness or receive information regarding possible prohibited protected characteristic discrimination or harassment, any form of sexual misconduct, and/or related retaliation, I am required to promptly report the information to the Office of Equity or their designee.

Religious Holiday Accommodation

Faculty in the University of Colorado system provide reasonable accommodations to students who must be absent from classes because of religious holidays. If you will miss class or graded assignments in order to observe religious holidays, you must contact me with all course conflicts by the end of the first week of classes.

Student Code of Conduct

As members of the University community, students are expected to uphold university standards, which include abiding by state civil and criminal laws and all University policies and standards of conduct. These standards are outlined in the student code of conduct, which can be found at <https://www.ucdenver.edu/student/health-wellness/student-conduct>

Academic Honesty

Students are expected to know, understand, and comply with the ethical standards of the university. A university's reputation is built on a standing tradition of excellence and scholastic integrity. As

members of the University of Colorado Denver academic community, faculty and students accept the responsibility to maintain the highest standards of intellectual honesty and ethical conduct.

Academic dishonesty is defined as a student's use of unauthorized assistance with intent to deceive an instructor or other such person who may be assigned to evaluate the student's work in meeting course and degree requirements.

Examples of academic dishonesty include, but are not limited to, the following:

Plagiarism. Plagiarism is the use of another person's distinctive words or ideas without acknowledgment. Examples include:

1. Word-for-word copying of another person's ideas or words;
2. The mosaic (the interspersing of one's own words here and there while, in essence, copying another's work);
3. The paraphrase (the rewriting of another's work, yet still using their fundamental idea or theory);
4. Fabrication of references (inventing or counterfeiting sources);
5. Submission of another's work as one's own;
6. Neglecting quotation marks on material that is otherwise acknowledged.

Acknowledgment is not necessary when the material used is common knowledge.

Cheating. Cheating involves the possession, communication, or use of information, materials, notes, study aids or other devices not authorized by the instructor in an academic exercise, or communication with another person during such an exercise. Examples include:

1. Copying from another's paper or receiving unauthorized assistance from another during an academic exercise or in the submission of academic material;
2. Using a calculator when its use has been disallowed;
3. Collaborating with another student or students during an academic exercise without the consent of the instructor.

Fabrication and Falsification. Fabrication involves inventing or counterfeiting information, i.e., creating results not obtained in a study or laboratory experiment. Falsification, on the other hand, involves the deliberate alteration of results to suit one's needs in an experiment or other academic exercise.

Misuse of Academic Materials. The misuse of academic materials includes, but is not limited to, the following:

1. Stealing or destroying library or reference materials or computer programs;
2. Stealing or destroying another student's notes or materials, or having such materials in one's possession without the owner's permission;
3. Receiving assistance in locating or using sources of information in an assignment when such assistance has been forbidden by the instructor;
4. Illegitimate possession, disposition, or use of examinations or answer keys to examinations;
5. Unauthorized alteration, forgery, or falsification;
6. Unauthorized sale or purchase of examinations, papers, or assignments.

Complicity in Academic Dishonesty. Complicity involves knowingly contributing to another's acts of academic dishonesty. Examples include:

1. Knowingly aiding another in any act of academic dishonesty;
2. Allowing another to copy from one's paper for an assignment or exam;
3. Distributing test questions or information about the materials to be tested before the scheduled exercise;
4. Taking an exam or test for someone else;
5. Signing another's name on attendance roster or on an academic exercise.

Incomplete Policy

When a student has special circumstances that make it impossible to complete course assignments, faculty members may choose to award an incomplete grade. All incomplete courses are assigned a grade of Incomplete (I). Incomplete grades are not awarded for poor academic performance or as a way of extending assignment deadlines. Faculty are not required to award an Incomplete.

To be eligible for an Incomplete grade, students MUST:

- Have participated in the class for a significant proportion of the term.
- Have successfully completed a significant proportion of the course assignments.
- Have special circumstances (verification may be required) that preclude the student from attending class and/or completing graded assignments.
- Make arrangements to complete missing assignments with the original instructor by a mutually agreed upon date but within one calendar year.
- Both the instructor and student should complete and sign the incomplete form found at

<https://clas.ucdenver.edu/faculty-staff/content/incomplete-grade-policy>.

- The instructor gives a copy of the signed incomplete form to the department.

Incompletes cannot:

- require a student to repeat the entire course,
- repeat or replace existing grades,
- allow the student an indeterminate period of time to complete a course, or
- allow the student to repeat the course with a different instructor.